

Skill 8 — Program 1: Environment Variable Expansion

Problem

Implement `$VARIABLE` expansion using `getenv()`.

```
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano skill8.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc skill8.c -o skill8
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./Skill8
-bash: ./Skill8: No such file or directory
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./skill8
Enter text: i love you moree
Expanded Output: i love you moree
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat skill8.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

int main()
{
    char input[200];
    char variable[100];

    int i = 0;
    int j = 0;

    printf("Enter text: ");
    fgets(input, sizeof(input), stdin);

    printf("Expanded Output: ");

    while (input[i] != '\0')
    {
        if (input[i] == '$')
        {
            i++;
            j = 0;

            while (input[i] != '\0' &&
                (isalnum((unsigned char)input[i]) ||
                 input[i] == '_'))
            {
                if (j < 99)
```

```
while (input[i] != '\0' &&
      (isalnum((unsigned char)input[i]) ||
       input[i] == '_'))
{
    if (j < 99)
    {
        variable[j++] = input[i];
    }

    i++;
}

variable[j] = '\0';

if (j > 0)
{
    char *value = getenv(variable);

    if (value != NULL)
    {
        printf("%s", value);
    }
    else
    {
        printf("$");
    }
}
else
{
    printf("%c", input[i]);
    i++;
}

return 0;
}

saikrishna_90156@DESKTOP-2JF9H5T:~$ |
```